COMP34212-Lab-S-Report

Dongmyung Park Student ID: 10879360

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1 Part 1 — Deep Neural Networks in Robotic Perception: State-of-the-Art

Deep Neural Networks (DNNs) are multi-layered computational models that learn hierarchical feature representations directly from raw data through end-to-end optimisation, eliminating the need for hand-crafted feature engineering that constrained classical robotics pipelines.

This capacity has made DNNs the dominant paradigm across virtually all subfields of cognitive robotics [4]. In robotic manipulation, supervised deep learning enables data-driven grasp quality prediction from depth images; learned grasp quality metrics have been shown to outperform hand-engineered heuristics on novel objects, yet require large synthetic training datasets and generalise poorly across fundamentally different object morphologies [4]. In unmanned aerial vehicles (UAVs), lightweight convolutional networks provide real-time obstacle avoidance under tight power budgets, though robustness to adverse weather and illumination changes remains an open challenge. In human-robot interaction (HRI), transformer-based architectures fuse speech and gesture streams to enable natural communication, though sensitivity to domain shift requires careful per-deployment fine-tuning. In surgical robotics, convolutional segmentation networks classify tissue types and track instruments directly from endoscopic video, where annotation scarcity and patient-specific variability fundamentally limit generalisation beyond controlled clinical settings.

Across all these domains a consistent pattern emerges: higher perceptual capability demands disproportionate computational cost. This tension is most acutely expressed in autonomous driving, which requires *simultaneous* perception, prediction of dynamic agent behaviour, and safe trajectory planning in real time on resource-constrained onboard hardware. A single 100 ms inference delay translates to nearly 3 metres of uncontrolled travel at 100 km/h [4], making latency a direct safety constraint rather than merely a performance metric. This section critically analyses three dominant DNN paradigms for autonomous driving and identifies their limitations, which directly motivate the Knowledge Distillation experiments in Part 2.

1.1 Planning-Oriented End-to-End Networks

End-to-End (E2E) driving models address the compounding error propagation of modular pipelines — where a misclassification in perception cascades into planning failures — by integrating all modules into a single unified DNN. UniAD [1] uses query-based Transformers to process raw sensor data and directly output control commands, achieving state-of-the-art performance on nuScenes and winning Best Paper at CVPR 2023 through holistic joint optimisation. However, two critical limitations constrain deployment. First, the black-box nature of deep Transformer stacks prevents localisation of failure root causes across perception, prediction, and planning, making regulatory compliance infeasible under automotive safety standards such as ISO 26262 — a fundamental barrier to commercial deployment, not merely a research limitation. Second, at 1.7 TFLOPs per forward pass on NVIDIA's automotive-grade Orin SoC ($\sim$275 TOPS) [1], prohibitive compute leaves insufficient headroom for concurrent safety-critical tasks such as emergency braking and multi-sensor fusion.

1.2 3D Occupancy Networks for Scene Representation

Traditional 3D bounding boxes cannot represent irregularly shaped obstacles such as construction debris or articulated vehicles. 3D Occupancy Networks [2] address this by dividing the surrounding space into dense voxel grids, using 3D CNNs or Transformers to predict per-voxel occupancy and semantic class, enabling reasoning over the true geometric volume of arbitrary obstacles. However, at $0.4\,\mathrm{m}^3$ voxel resolution a typical 200m $\times$ 200m $\times$ 8m driving scene yields $\sim$5 million voxels ($500 \times 500 \times 20$), simultaneously imposing GPU memory demands incompatible with the 8–16 GB VRAM of automotive processors and preventing sub-100 ms inference — a hard real-time constraint that this representation structurally cannot satisfy without lossy approximations that undermine its core geometric accuracy advantage.

1.3 Cognitive Reasoning via Large Vision-Language Models

To address long-tail scenarios that deterministic pipelines cannot anticipate — such as a pedestrian chasing a ball onto the road — the latest trend integrates Large Vision-Language Models (LVLMs). Models such as DriveVLM [3] process visual inputs alongside natural language reasoning chains, enabling cognitive scene understanding analogous to human situational awareness. Despite compelling generalisation to novel scenarios, three deployment barriers remain: the $\sim$19 B parameter architecture (ViT-6B visual encoder plus Vicuna-13B language model) is fundamentally incompatible with real-time latency requirements; hallucinated scene

descriptions can trigger unsafe planning decisions; and the absence of formal verification guarantees means safety certification under existing regulatory frameworks remains an open, unsolved challenge [3].

1.4 Summary and Rationale for the Proposed Methodology

Table 1: Comparative summary of three state-of-the-art DNN paradigms in autonomous driving, illustrating the capability–efficiency tension that motivates model compression.

Architecture	Key Strength	Primary Limitation	Edge Viable
UniAD (E2E) [1]	Joint end-to-end optimisation; SOTA nuScenes	Black-box; prohibitive FLOPs; no safety cert.	No
Occ3D [2]	Arbitrary obstacle geometry; semantic voxels	∼5M voxels; VRAM >16 GB; >100 ms latency	No
DriveVLM [3]	Long-tail generalisation; language reasoning	Hallucination; ∼19 B params; no formal verification	No

Table 1 reveals that every advance in perceptual capability introduces a proportional increase in computational cost that renders edge deployment infeasible. All three paradigms converge on the same barrier: architectures too demanding for resource-constrained onboard processors where real-time inference and energy efficiency are non-negotiable [4]. Critically, this capability–efficiency tension is not unique to autonomous driving — the same pattern characterises manipulation, HRI, UAV, and surgical robotics, confirming model compression as a universal and unsolved challenge in cognitive robotics. Part 2 directly addresses this by evaluating Knowledge Distillation (KD) as a principled strategy to compress a capable teacher model into a lightweight edge-deployable student, with rigorous quantification of the resulting accuracy–efficiency trade-off.

2 Part 2 — Knowledge Distillation for Edge-Deployed Traffic Sign Recognition

2.1 Methodology

2.1.1 Problem Formulation and Dataset

This experiment evaluates KD as a model compression strategy for traffic sign recognition under edge deployment constraints [4]. KD was selected over pruning and quantisation because it requires no post-hoc architectural modification and imposes zero inference overhead: the Student architecture remains fixed at deployment, making it directly compatible with resource-constrained hardware [7]. Unlike pruning (iterative mask search, accuracy recovery cycles) or quantisation (hardware-specific integer arithmetic constraints), KD transfers knowledge purely through the training loss signal — the most deployment-transparent compression method available, and the one most directly motivated by the edge deployment barriers identified in Part 1.

The GTSRB dataset [6] (43 classes, 39,209 training / 12,630 test images) was selected for its direct correspondence to the autonomous driving perception task — traffic sign recognition is a safety-critical sub-component of every production driving system — and its natural class imbalance reflecting realistic deployment distributions. Images were resized to 96×96, normalised to $[0, 1]$, with inverse-frequency class weights and fixed seed (10879360). Code: `https://www.kaggle.com/code/dongmyungpark/gtsrb`.

2.1.2 Model Architectures and Computational Complexity

Teacher Model. A MobileNetV2 backbone [5] (ImageNet pre-trained, top 60 layers fine-tuned) with GlobalAveragePooling2D, Dropout(0.3), and Dense(43), yielding 2,313,067 parameters and ∼300 MFLOPs at 96×96. Safe augmentation applied (RandomBrightness/Contrast ±20%, RandomZoom ±5%; rotation excluded to preserve direction-sensitive signs). Adam (lr=1×10^{-5}, ReduceLROnPlateau factor = 0.5, patience = 3), 15 epochs: **88.37%** test accuracy (validation: 88.99%). This reflects an intentionally constrained training regime mirroring real-world resource-limited conditions rather than the architecture's ceiling; published GTSRB state-of-the-art with full optimisation exceeds 99% [6]. This conservative setup establishes a realistic teacher whose soft labels remain informative without requiring prohibitive training compute, accurately representing resource-limited deployment scenarios.

Student Baseline Model. A custom MicroCNN (34,406 parameters; 67× fewer) is built ground-up around the depthwise-separable convolution philosophy of MobileNetV2 [5]: three SeparableConv2D blocks (32/64/128 filters), BatchNorm, MaxPooling, GlobalAveragePooling2D, Dense(128), and Dropout(0.2). SeparableConv2D factorises $3{\times}3{\times}C$ convolutions into depthwise and pointwise steps, yielding ∼6 MFLOPs (50× reduction vs. Teacher) while preserving spatial feature extraction capacity — directly simulating the compute profile of onboard processors such as the Jetson Nano [4]. Trained for 10 epochs without knowledge transfer: 54.61% test accuracy (performance lower bound).

2.1.3 Knowledge Distillation Setup

Offline KD was applied following Hinton et al. [7], wherein the frozen Teacher generates soft probability distributions over all 43 classes, encoding inter-class visual similarities absent from one-hot labels. The

training loss combines hard and soft components:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{hard}} + (1-\alpha) \cdot T^2 \cdot \mathcal{L}_{\text{soft}} \tag{1}$$

where $\mathcal{L}_{\text{hard}}$ is SparseCategoricalCrossentropy, $\mathcal{L}_{\text{soft}}$ is the KL divergence between temperature-softened logits ($\text{softmax}(\mathbf{z}/T)$), and T^2 compensates for gradient suppression from temperature scaling [7]. Higher T produces softer distributions that reveal more inter-class similarity, while lower T preserves sharper discriminative signal. A 3×3 grid search over $T \in \{3, 5, 10\}$ and $\alpha \in \{0.1, 0.3, 0.5\}$ (9 conditions) captures joint hyperparameter effects, analysed along **Experiment A** (α=0.1 fixed, T swept) and **Experiment B** (T=10 fixed, α swept).

2.2 Results and Analysis

2.2.1 Accuracy–Efficiency Trade-off

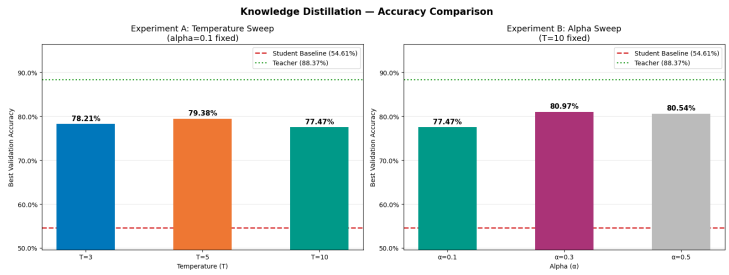
Model Comparison: Teacher vs Student (GTSRB, 10 Epochs)
★ Best KD condition: T=10_a=0.3 (80.97%)

Model	Parameters	Latency (ms/img)	Val Accuracy	vs Baseline
Teacher (MobileNetV2)	2,313,067	—	88.37%	—
Student Baseline	34,406	54.01 ± 2.96 ms	54.61%	0.00 pp
KD T=3 α=0.1	34,406	53.73 ± 2.22 ms	78.21%	+23.60 pp
KD T=3 α=0.3	34,406	53.68 ± 2.46 ms	76.43%	+21.82 pp
KD T=3 α=0.5	34,406	54.82 ± 3.48 ms	76.66%	+22.05 pp
KD T=5 α=0.1	34,406	53.74 ± 2.08 ms	79.38%	+24.77 pp
KD T=5 α=0.3	34,406	55.03 ± 3.42 ms	80.29%	+25.68 pp
KD T=5 α=0.5	34,406	53.76 ± 2.28 ms	76.98%	+22.37 pp
KD T=10 α=0.1	34,406	53.91 ± 2.37 ms	77.47%	+22.86 pp
KD T=10 α=0.3	**34,406**	**55.19 ± 3.79 ms**	**80.97%**	**+26.37 pp**
KD T=10 α=0.5	34,406	54.44 ± 2.41 ms	80.54%	+25.93 pp

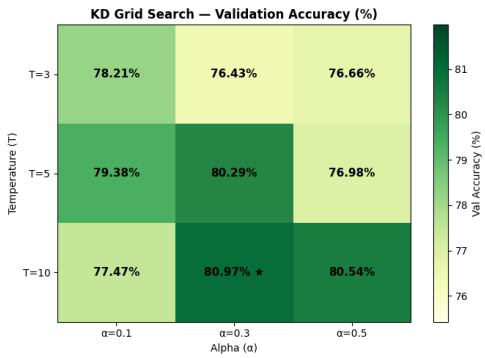
Figure 1: Model comparison across all 9 KD conditions: parameter count, inference latency (mean±std, 100 runs, batch = 1), and validation accuracy. Best KD condition (T=10, α=0.3) highlighted in gold.

Figure 1 summarises all conditions on the held-out GTSRB *test* set. Teacher: 88.37% (2,313,067 params); Student Baseline: 54.61% (34,406 params) — a 33.76 pp gap confirming that a 67× parameter and 50× FLOPs reduction substantially limits hard-label learning. Crucially, all Student models achieve near-identical inference latency (53.68–55.19 ms/img, spread only 1.51 ms across all 9 conditions), confirming that KD hyperparameter choice introduces zero inference overhead at deployment — the key practical advantage over pruning and quantisation.

2.2.2 Knowledge Distillation Results



(a) Bar charts: Exp. A (Temperature Sweep) and Exp. B (Alpha Sweep). Dashed red: Baseline (54.61%). Dotted green: Teacher (88.37%).



(b) Full 3×3 grid search heatmap ($T \times \alpha$). Best cell: T=10, α=0.3 at 80.97%.

Figure 2: KD hyperparameter sweep results. (a) Marginal effects of T and α. (b) Full grid reveals that peak performance requires joint tuning: the optimal α shifts depending on T.

As shown in Figure 2, KD consistently improved Student accuracy across all 9 conditions, with every condition exceeding the Baseline by at least +21.8 pp, confirming that soft-label transfer provides robust learning signal regardless of hyperparameter configuration. The best condition, KD (T=10, α=0.3), achieved **80.97%** — 91.6% of Teacher accuracy, recovering 78.1% of the Teacher–Baseline gap ($\frac{80.97-54.61}{88.37-54.61}$) at 67× fewer parameters and identical inference cost. This is a direct empirical answer to the edge deployment challenge identified in Part 1: KD compresses a capable model to the Jetson Nano compute profile while retaining the overwhelming majority of its task accuracy.

2.2.3 Hyperparameter Analysis: Temperature and Alpha Interaction

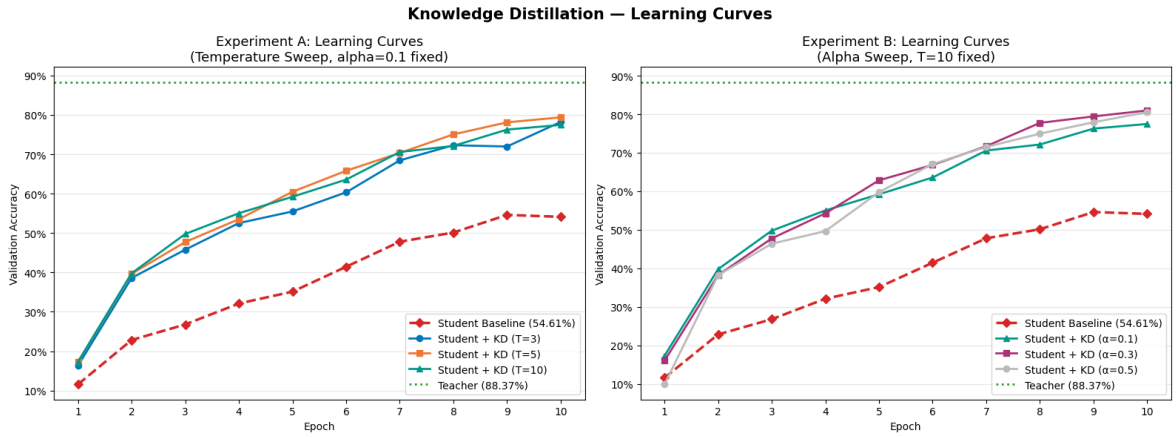
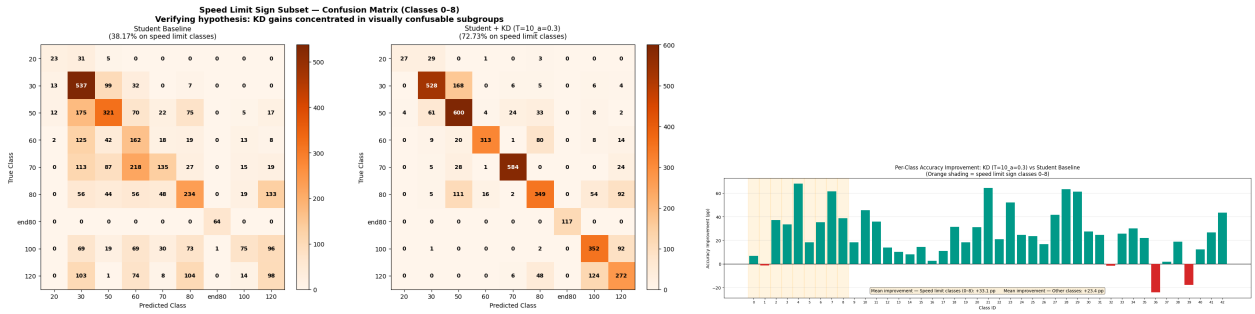


Figure 3: Epoch-by-epoch learning curves. Left: Experiment A (Temperature Sweep, α=0.1 fixed). Right: Experiment B (Alpha Sweep, T=10 fixed). All KD conditions consistently outperform the Baseline from Epoch 1, confirming that soft-label guidance accelerates early feature learning regardless of hyperparameter choice.

Experiment A (Temperature Sweep, α=0.1 fixed). A non-monotonic response emerges with a *peak* at T=5: T=5 (79.38%) > T=3 (78.21%) > T=10 (77.47%). All KD conditions outperform the Baseline from Epoch 1 (Figure 3), confirming that soft-label guidance accelerates convergence from the earliest training steps — a property of particular value in resource-constrained deployments where training epochs are limited. At T=10 with α=0.1, excessively flat soft distributions dilute the hard-label signal. The full grid (Figure 2b) reveals this as an artefact of the fixed-α constraint: jointly tuning to α=0.3 unlocks T=10's peak at 80.97%, while T=5, α=0.3 achieves a competitive 80.29%, confirming that high-temperature benefits require appropriate hard-label grounding.

Experiment B (Alpha Sweep, T=10 fixed). An inverted-U emerges: α=0.3 (80.97%) > α=0.5 (80.54%) > α=0.1 (77.47%). α=0.1 under-weights hard labels given the very flat T=10 distributions, while α=0.5 begins to discard the inter-class similarity encoded in the soft signal. This T–α interaction confirms that the optimal α is conditional on T: joint optimisation is necessary, as independent univariate tuning would have selected T=5, α=0.1 at only 79.38%, missing the 80.97% optimum by 1.59 pp. This finding has a direct practical implication: grid search over the joint (T, α) space is essential and cannot be replaced by sequential tuning.

2.2.4 Broad Efficacy and Per-Class Regression Analysis



(a) Speed limit subset (Classes 0–8). Baseline: 38.17% → KD: 72.73%.

(b) Per-class improvement. Speed limit mean: +33.1 pp vs. others: +23.4 pp.

Figure 4: Confusion matrix analysis. (a) Speed limit subset confirms the inter-class similarity hypothesis. (b) Per-class improvement: KD broadly effective across 43 classes with localised regressions in specific categories.

Figure 4(b) confirms broad KD efficacy across the 43 classes, with improvement concentrated in structurally similar sign groups. Two distinct transfer patterns emerge, providing mechanistic insight into how soft labels interact with visual class structure.

Positive transfer — semantically coherent similarity. Speed limit signs (Classes 0–8) share identical circular red borders differing only in numeral. The Baseline achieved only 38.17% on this subset due to feature confusion between nearly identical templates; the KD Student improved to 72.73% (+34.56 pp). Mean per-class improvement on speed limit classes (+33.1 pp) substantially exceeds other classes (+23.4 pp) — direct evidence that T=10 soft labels encode rich inter-class similarity providing calibrated uncertainty over visually similar neighbours, rather than a binary target that cannot capture shared visual structure.

Negative transfer — semantically misleading similarity. Classes 36 (*Go Straight or Right*; ≈−25 pp)

and 39 (*Keep Left*; ≈ -17 pp) exhibit the most severe regressions. These directional signs share template-level similarity with adjacent categories, but unlike speed limit signs where visual neighbour similarity is semantically coherent (adjacent speeds are related concepts), directional cross-confusion is semantically misleading: a "go straight" sign and a "keep left" sign require opposite driving actions, so soft-label probability mass on the wrong neighbour actively degrades discriminability. This reveals a fundamental condition for KD efficacy: soft labels improve performance when inter-class visual similarity reflects structural semantic regularity, but introduce negative transfer when template-level similarity is semantically arbitrary.

2.3 Discussion and Future Work

The results demonstrate KD as a broadly effective, deployment-compatible compression strategy. The best Student (T=10, α=0.3) achieves 80.97% — 91.6% of Teacher accuracy — at 67× fewer parameters and identical latency (∼54 ms/image), recovering 78.1% of the compression-induced accuracy gap. Both models were trained under constrained compute budgets, so reported accuracies are conservative lower bounds; extended training would yield higher absolute performance while the relative KD gain is expected to persist, as the soft-label transfer mechanism is independent of training duration. The 3×3 grid search reveals a critical T–α interaction: joint tuning is essential, and all KD conditions outperform the Baseline from Epoch 1 (Figure 3), demonstrating accelerated feature learning particularly valuable in time-limited deployment scenarios.

Three key extensions are proposed: *(i)* **Extended training** beyond 15/10 epochs to quantify the full accuracy ceiling and verify stability of the optimal (T=10, α=0.3) configuration; *(ii)* **Selective soft-label masking** via class-conditional temperature scheduling — applying lower T to semantically ambiguous sign categories (e.g., Classes 36, 39) while maintaining high T for structurally coherent groups (speed limits) — to mitigate negative transfer without sacrificing positive transfer gains; *(iii)* **Pareto frontier analysis** across student sizes (10K–100K parameters) to systematically identify minimum viable architectures for different edge hardware tiers, from Raspberry Pi to Jetson Orin NX, enabling principled hardware-aware model selection.

3 Conclusion

Three state-of-the-art autonomous driving architectures — E2E Transformers, 3D Occupancy Networks, and LVLMs — share a common deployment barrier: computational cost incompatible with onboard hardware where latency directly governs safety. This capability–efficiency tension is universal across cognitive robotics, from manipulation to surgical systems, confirming model compression as a fundamental rather than domain-specific challenge. Part 2 demonstrated Knowledge Distillation as a practical and deployment-transparent solution: the best Student (T=10, α=0.3) achieved 80.97% — 91.6% of Teacher performance — at identical latency (∼54 ms/image) with 67× fewer parameters, and zero inference overhead relative to the uncompressed student. Crucially, T and α must be jointly optimised: independent tuning misses the global optimum by 1.59 pp. Per-class analysis revealed a mechanistic principle underlying KD efficacy: soft labels improve performance when inter-class visual similarity is semantically coherent (speed limits: +33.1 pp) but introduce negative transfer when template similarity is semantically misleading (Classes 36, 39). This insight — that KD performance is determined by the alignment between visual and semantic class structure — directly informs future compression strategies for safety-critical robotic perception tasks where class boundaries must be both visually and semantically well-defined.

References

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